

Assignment 1

1 Problem 1.1

1.1 Problem 1.1.1

Note that $X \in \mathbb{R}^{d \times h}$, $w \in \mathbb{R}^{d \times 1}$, we have

$$X^T w = \begin{bmatrix} x_{1,1}w_1 + x_{2,1}w_2 + \cdots + x_{d,1}w_d \\ x_{1,2}w_1 + x_{2,2}w_2 + \cdots + x_{d,2}w_d \\ \vdots \\ x_{1,h}w_1 + x_{2,h}w_2 + \cdots + x_{d,h}w_d \end{bmatrix} = \begin{bmatrix} f_1 \\ f_2 \\ \vdots \\ f_h \end{bmatrix}$$

$$\frac{d(X^T w)}{dw} = \begin{bmatrix} \frac{\partial f_1}{\partial w_1} & \cdots & \frac{\partial f_h}{\partial w_1} \\ \vdots & \ddots & \vdots \\ \frac{\partial f_1}{\partial w_d} & \cdots & \frac{\partial f_h}{\partial w_d} \end{bmatrix} = \begin{bmatrix} x_{1,1} & \cdots & x_{1,h} \\ \vdots & \ddots & \vdots \\ x_{d,1} & \cdots & x_{d,d} \end{bmatrix} = X$$

Remark: In this problem, X cannot be treated as a vector, so the dot product is unavailable. It should be solved by differentiation of a vector function w.r.t. a vector.

1.2 Problem 1.1.2

Note that $X \in \mathbb{R}^{h \times d}$ is a matrix, $w \in \mathbb{R}^{d \times 1}$ and $y \in \mathbb{R}^{h \times 1}$ are vectors. We have

$$\frac{d(y^T X w)}{dw} = \frac{d((y^T X) w)}{dw} = (y^T X)^T = X^T y$$

Alternative Methods:

Write down the scalar form of $y^T X w$ and calculate the derivatives by using differentiation of a scalar function w.r.t. a vector.

1.3 Problem 1.1.3

Note that $X \in \mathbb{R}^{d \times d}$ and $w \in \mathbb{R}^{d \times 1}$, we have

$$\begin{aligned}
w^T X w &= w_1 (x_{1,1}w_1 + x_{1,2}w_2 + \cdots + x_{1,d}w_d) \\
&+ w_2 (x_{2,1}w_1 + x_{2,2}w_2 + \cdots + x_{2,d}w_d) \\
&+ \cdots \\
&+ w_d (x_{d,1}w_1 + x_{d,2}w_2 + \cdots + x_{d,d}w_d).
\end{aligned}$$

Define $f = w^T X w$, we have

$$\frac{d(w^T X w)}{dw} = \begin{bmatrix} \frac{\partial f}{\partial w_1} \\ \vdots \\ \frac{\partial f}{\partial w_d} \end{bmatrix} = \begin{bmatrix} x_{1,1}w_1 + \cdots + x_{1,d}w_d + x_{2,1}w_2 + \cdots + x_{d,1}w_d \\ x_{2,1}w_1 + \cdots + x_{2,d}w_d + x_{1,2}w_1 + \cdots + x_{d,2}w_d \\ \vdots \\ x_{d,1}w_1 + \cdots + x_{d,d}w_d + x_{1,d}w_1 + \cdots + x_{d,d}w_d \end{bmatrix} = (X + X^T) w$$

Alternative Methods:

Use the rule of differential of composite function $\frac{d(u^T v)}{dw} = \frac{du^T}{dw} v + \frac{dv^T}{dw} u$. (You can find more details in Matrix Cookbook [1] Chapter 2.4).

Define $u = w$ and $v = Xw$, we have

$$\begin{aligned}
\frac{d(w^T X w)}{dw} &= \frac{d(u^T v)}{dw} \\
&= \frac{du^T}{dw} v + \frac{dv^T}{dw} u \\
&= \frac{dw^T}{dw} X w + \frac{d(w^T X^T)}{dw} w \\
&= X w + X^T w \\
&= (X + X^T) w,
\end{aligned}$$

where I is identity matrix.

2 Problem 1.2

2.1 Problem 1.2.1

Set

$$\mathbf{W} = \begin{bmatrix} \mathbf{b}^T \\ \mathbf{w}^T \end{bmatrix} \in \mathbb{R}^{(d+1) \times k} \quad \mathbf{X} = \begin{bmatrix} 1 & \mathbf{x}_1^T \\ \vdots & \vdots \\ 1 & \mathbf{x}_N^T \end{bmatrix} \in \mathbb{R}^{N \times (d+1)} \quad \mathbf{Y} = \begin{bmatrix} \mathbf{y}_1^T \\ \vdots \\ \mathbf{y}_N^T \end{bmatrix} \in \mathbb{R}^{N \times k}$$

$$\mathbf{A} = \begin{bmatrix} \alpha_1 & & \\ & \ddots & \\ & & \alpha_N \end{bmatrix} \in \mathbb{R}^{N \times N}$$

Therefore, we have

$$\begin{aligned}
J(\mathbf{W}) &= J(\mathbf{w}, \mathbf{b}) \\
&= \sum_{i=1}^N \alpha_i \|\mathbf{y}_i - \mathbf{W}\mathbf{x}_i - \mathbf{b}\|^2 \\
&= \text{trace} [(\mathbf{Y} - \mathbf{X}\mathbf{W})^T \mathbf{A} (\mathbf{Y} - \mathbf{X}\mathbf{W})] \\
&= \text{trace} [\mathbf{Y}^T \mathbf{A} \mathbf{Y} - \mathbf{W}^T \mathbf{X}^T \mathbf{A} \mathbf{Y} - \mathbf{Y}^T \mathbf{A} \mathbf{X} \mathbf{W} + \mathbf{W}^T \mathbf{X}^T \mathbf{A} \mathbf{X} \mathbf{W}]
\end{aligned}$$

We can calculate the derivation of $J(\cdot)$ with respect to $\mathbf{W}$ as

$$\begin{aligned}
\frac{\partial J(\mathbf{W})}{\partial \mathbf{W}} &= -\mathbf{X}^T \mathbf{A} \mathbf{Y} - \mathbf{X}^T \mathbf{A}^T \mathbf{Y} + (\mathbf{X}^T \mathbf{A} \mathbf{X} + \mathbf{X}^T \mathbf{A} \mathbf{X}) \mathbf{W} \\
&= -2\mathbf{X}^T \mathbf{A} \mathbf{Y} + 2\mathbf{X}^T \mathbf{A} \mathbf{X} \mathbf{W}
\end{aligned}$$

Let $\frac{\partial J(\mathbf{W})}{\partial \mathbf{W}} = 0$, we have

$$\mathbf{W}^* = (\mathbf{X}^T \mathbf{A} \mathbf{X})^{-1} \mathbf{X}^T \mathbf{A} \mathbf{Y},$$

where $\mathbf{X}^T \mathbf{A} \mathbf{X}$ is invertible.

2.2 Problem 1.2.2

1. Initialize: Choose learning rate β and iteration T , and initialize $t = 0$ and $\mathbf{W}_0$.
2. Update parameters $\mathbf{W}$ with gradient descent

$$\mathbf{W}_{t+1} = \mathbf{W}_t - \beta \frac{\partial J(\mathbf{W})}{\partial \mathbf{W}}$$

3. Repeat the last step for T -times iteration untill convergence.
4. The final parameters are $\mathbf{W}_T$.

3 Problem 1.3

3.1 Problem 1.3.1

Calculating the derivative of f with respect to x , we have

$$f'(x) = 4x^3,$$

$$f''(x) = 12x^2.$$

Obviously, $\forall x \in \mathbb{R}, f''(x) \geq 0$. Therefore, $f(x)$ is convex.

3.2 Problem 1.3.2

Suppose that $\theta \in (0, 1)$, $x, y \in \mathbb{R}$

$$\begin{aligned} f(\theta x + (1 - \theta)y) &= |\theta x + (1 - \theta)y| \\ &\leq |\theta x| + |(1 - \theta)y| \\ &= \theta|x| + (1 - \theta)|y| \\ &= \theta f(x) + (1 - \theta)f(y), \end{aligned}$$

i.e. $f(\theta x + (1 - \theta)y) \leq \theta f(x) + (1 - \theta)f(y)$. Therefore, $f(x)$ is convex.

3.3 Problem 1.3.3

$$\begin{aligned} f(\mathbf{x}) &= \|\mathbf{Ax} - \mathbf{b}\|^2 \\ &= (\mathbf{Ax} - \mathbf{b})^T (\mathbf{Ax} - \mathbf{b}) \\ &= \mathbf{x}^T \mathbf{A}^T \mathbf{Ax} - 2\mathbf{b}^T \mathbf{Ax} + \mathbf{b}^T \mathbf{b}. \end{aligned}$$

We can calculate its derivation as

$$\frac{\partial f(\mathbf{x})}{\partial \mathbf{x}} = 2\mathbf{A}^T \mathbf{Ax} - 2\mathbf{A}^T \mathbf{b},$$

$$\frac{\partial^2 f(\mathbf{x})}{\partial \mathbf{x} \partial \mathbf{x}} = 2\mathbf{A}^T \mathbf{A}.$$

Obviously, $\forall \mathbf{x} \in \mathbb{R}^n$, $\frac{\partial^2 f(\mathbf{x})}{\partial \mathbf{x} \partial \mathbf{x}} \geq 0$. Therefore, $f(\mathbf{x})$ is convex.

4 Problem 1.4

First, write down the p.d.f. of Normal Distribution as

$$\begin{aligned} L(\mu, \sigma \mid x_1 \cdots x_N) &= \prod_{i=1}^N \left(\frac{1}{\sqrt{2\pi\sigma^2}} \exp \frac{-(x_i - \mu)^2}{2\sigma^2} \right) \\ &= \frac{1}{\sqrt{(2\pi\sigma^2)^N}} \exp \left(-\frac{1}{2\sigma^2} \sum_{i=1}^N (x_i - \mu)^2 \right) \end{aligned}$$

Then take the log of both sides, we have

$$\ln L(\mu, \sigma \mid x_1 \cdots x_N) = -\frac{N}{2} (\ln 2\pi + \ln \sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^N (x_i - \mu)^2$$

Then let the derivatives equal to 0, we have

$$\begin{aligned}\frac{\partial}{\partial \mu} \ln L(\mu, \sigma \mid x_1 \cdots x_N) &= \frac{1}{\sigma^2} \sum_{i=1}^N (x_i - \mu) \\ &= 0, \\ \frac{\partial}{\partial \sigma^2} \ln L(\mu, \sigma \mid x_1 \cdots x_N) &= -\frac{N}{2\sigma^2} + \frac{1}{2(\sigma^2)^2} \sum_{i=1}^N (x_i - \mu)^2 \\ &= 0.\end{aligned}$$

Here, for the Normal(Gaussian) distribution, $x_1 \cdots x_N$ are i.i.d sampled from $\mathbf{N}(\mu, \sigma^2)$, the log-likelihood function is concave [2], hence the second order sufficient condition can be satisfied and the solution are global.

Finally, we have $\sigma_{ML}^2 = \frac{1}{N} \sum_{i=1}^N (x_i - \mu_{ML})^2$, where $\mu_{ML} = \bar{x}$.

References

- [1] Kaare Brandt Petersen, Michael Syskind Pedersen, et al. The matrix cookbook. *Technical University of Denmark*, 7(15):510, 2008.
- [2] John W Pratt. Concavity of the log likelihood. *Journal of the American Statistical Association*, 76(373):103–106, 1981.